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Music Genre Classification Using Logistic Regression, K-Nearest Neighbors, and PCA-Enhanced KNN

Historically, music genre classification has been a problem in the world of music information retrieval. The task is to assign a genre label to a song using information extracted from the audio signal. Accurate genre classification can support music recommendation, playlist generation, organization of digital libraries, and broader content-based retrieval tasks. Even from a machine learning perspective, the problem is an interesting one with regard to musical genres as there are so many songs out there that cross genre boundaries. This makes classification rather noisy and prone to imperfection.

The aim of this project's study is to use the public dataset provided by Free Music Archive and predicting each track's top-level genre using precomputed numerical audio features. Instead of training on the raw audio directly and consuming hours of time to do so, the project uses the official metadata and feature tables provided with FMA. This method makes it possible to concentrate on model comparison and experimental design. I explored three different methods of examination in my research. The first was multinomial logistic regression as a baseline for my research. The second method for comparison was k-nearest neighbors to provide a distance-based non-parametric method of examination. Finally, my third approach was a modified hybrid approach that applied Principal Component Analysis before KNN in order to reduce dimensionality and possibly improve neighbor-based classification in a high-dimensional feature space like the FMA metadata. The PCA+KNN model served as my proposed improved method.

The data used for the project, as aforementioned, came from the official FMA metadata package. In particular, the project relied heavily on the tracks.csv, genres.csv and features.csv. The tracks.csv contained track-level metadata, to include top-level genre labels. The features.csv contained precomputed numeric descriptors for each of those tracks. The features provided an easy starting point for the machine learning experimentation without the need to process all of the raw audio ahead of time. First, the top-level genre label was extracted from the track/genre_top field and aligned with the feature table using the track ID index. After removing tracks with missing top-level genre labels, what was left were 49,598 tracks and 518 numerical feature columns.

To make this experiment more realistic, I chose to restrict my analysis to eight top level genres that included Hip-Hop, Pop, Rock, Experimental, Folk, Jazz, Electronic, and Soul-RnB. I chose these labels as they provided a nice meaningful mix of classes ranging from high-frequency to low-frequency. After filtering the full dataset to just these eight genres I was left with 43,595 in my dataset. I, however, ran into another issue in regards to distribution imbalance. The largest classes, that of Rock, Experimental and Electronic contained far more examples than Jazz and Soul-RnB. To even out this distribution and reduce the dominance of the former large class of genres, the larger genres were down sampled to 1000 tracks per class. Doing so produced a final working dataset of 6,746 tracks and 518 features. This reduced dataset now included 1000 tracks from Hip-Hop, Pop, Rock, Experimental, Folk and Electronic, along with 571 Jazz tracks and 175 Soul-RnB tracks.

Following distribution, the dataset was then split into training and test sets in an 80/20 split so that the genre proportions would be preserved across both ends of the split. The results of this split was 5,396 training examples and 1,350 test examples. The StandardScaler became essential during this phase as both logistic regression and KNN are sensitive to scaling in their own unique ways. KNN's distance-based classifiers can be distorted when some variables have much larger

numeric ranges, for example. The use of scikit-learn pipelines also ensured that preprocessing and classification were applied with consistency during fitting and evaluation.

The first model method experimented with was the multinomial logistic regression. Logistic regression is a strong baseline as a linear classifier as it estimates class probabilities as a function of weighted input features. The output of the method, then, is simple, interpretable and often effective when the classes are reasonably separable in a linear feature space. To ensure good convergence, the logistic regression method was trained with standardized features, a multinomial objective and a sufficiently high iteration limit. On this held-out test set, this method ultimately achieved an accuracy of approximately 0.5319, or 53.2%.

The second model method experimented with was the k-nearest neighbors method. KNN does not learn a global set of model coefficients as in the case of logistic regression; however, when the true class boundary is nonlinear (as can be the case with many genres of music) making decisions locally, based on neighborhood structure can be useful. However, KNN can struggle when the data is high-dimensional or noisy. To explore KNN's efficacy, a test was done with $k = 3, 5, 7$ and 9 . The best result occurred at $k = 7$ which produced an accuracy of approximately 0.5055, or 50.4%. Smaller values like $k = 3$ performed much worse. This suggested that very local neighborhoods may have been more sensitive to noise and instability in this particular space.

The third method was designed to be a modified version for KNN. Since the dimensionality of the dataset was relatively high with 518 features, the thought process was that a pure nearest-neighbor approach would not be as efficacious. High-dimensional settings can make distance measures less informative. To address this, PCA was added before KNN in an effort to reduce dimensionality. The intuition behind this method was that the lower-dimensional representation might preserve the most informative variation of the data while removing some of the noise, thus

making nearest-neighbor comparisons a little more meaningful. This produced a hybrid pipeline consisting of standardization, PCA and KNN. PCA was tested in dimensions of 20, 50 and 100, each with $k = 3, 5$ and 7 .

The best of the PCA+KNN configurations was 100 components with $k = 7$. The test accuracy for the experiment was approximately 0.5044, which essentially matched the best result from the standard KNN. Lower dimensions performed worse overall with 20-components showing the weakest performance. This would suggest that reducing the dimensions aggressively removed too much class-relevant information, while having 100-component retained just the right amount of structure to recover the same performance that I had with KNN on its own. Even so, through all of this testing, PCA+KNN, despite my initial hypothesis, did not produce a clear improvement over the strongest baseline KNN model.

Taken together, the results show a modest but consistent ranking among the methods. Logistic regression performed the best overall with 53.2% accuracy, while both the KNN and my proposed PCA+KNN hybrid reached about 50.4%. The outcome of these experiments show that the FMA feature representation used here is better suited to a global linear model than to a purely local distance-based methodology, likely due to the 518-dimensional features already separating the broad genre differences reasonably well in linear decision boundaries. KNN remained more vulnerable to uneven class density and residual noise in higher dimensional spaces.

Even though PCA+KNN did not surpass the standard KNN as I had originally hoped, the hybrid experiment is still rather informative. It was a motivated attempt to mitigate the problem of dimensionality by compressing the feature space before applying a nearest-neighbor classifier. The failure in the effort to improve performance suggests that dimensionality on its own is not the main bottleneck for KNN, particularly as it relates to this subset of data. The project also incorporated

several smaller but meaningful adaptations such as focusing on a subset of the top-level genres, down sampling majority classes, standardizing features within pipelines, and exploring multiple hyperparameter settings. The choices made in the experimentation, along with the underwhelming result for PCA+KNN, provide a clear understanding of how linear, local, and dimension-reducing methods behave as it relates to music genre classification and set the stage for future work with richer models and more detailed evaluation.